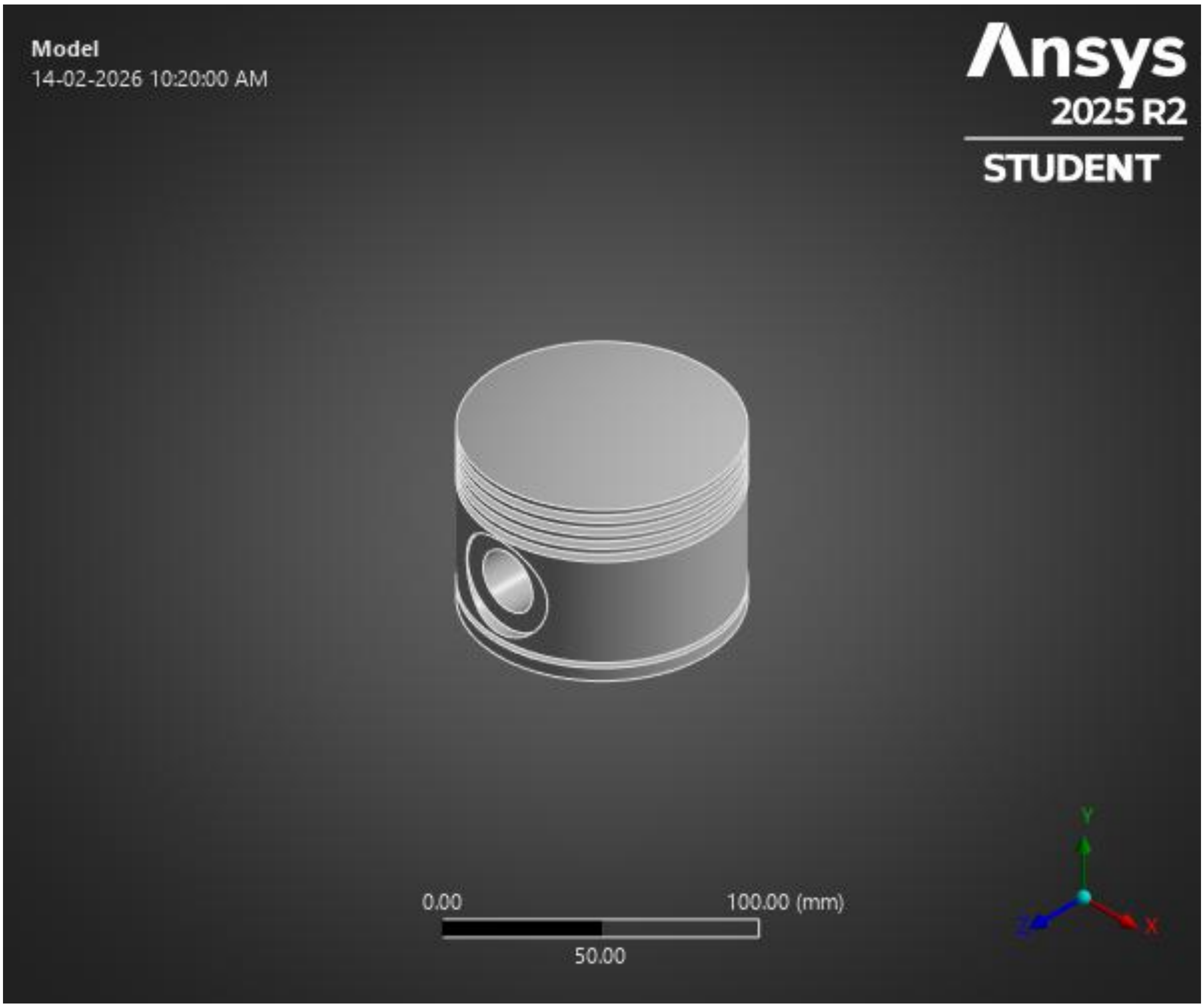
Finite Element Analysis & Design Validation

Engine – Multi Component Study



Prepared By:
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Design Engineer

Software Used:
SolidWorks *(3D Modelling)*
ANSYS Workbench 2025 R2 *(FEA & Modal Analysis)*

Date:
February 2026

Project Type:
Structural • Thermal • Fatigue • Modal Analysis

Objective

The objective of this study was to evaluate the structural integrity, thermal behaviour, fatigue durability, and dynamic stability of key components in a radial engine assembly using Finite Element Analysis (FEA).

The components analyzed include:

- Piston (Structural Validation)
- Cylinder / Barrel (Thermal & Thermal-Structural Coupling)
- Connecting Rod (Fatigue & Durability Assessment)
- Crankshaft (Modal & Dynamic Stability Analysis)

The study aims to verify whether the current design operates safely under assumed peak combustion and operating conditions.

Methodology

The 3D models were developed in SolidWorks and imported into ANSYS Workbench 2025 R2 for analysis.

The following simulations were performed:

- Static Structural Analysis
- Steady-State Thermal Analysis
- Thermal-Structural Load Transfer
- Fatigue Life Prediction (S–N approach)
- Modal Analysis (Eigenvalue extraction)

Peak combustion pressure of 6 MPa was assumed, resulting in an approximate piston load of 38 kN.
Engine operating speed was assumed at 2500 RPM (41.67 Hz).

Key Findings

Piston – Structural Performance

- Maximum Stress: 92 MPa
- Yield Strength: 280 MPa
- Safety Factor: 2.8
- Maximum Deformation: 0.141 mm

Stress levels remain well below yield limits, confirming structural safety.

Cylinder – Thermal Stability

- Maximum Thermal Deformation: **0.2386 mm**
- Thermal Strain: **1.77×10^{-3}**
- 100% load mapping during thermal-structural coupling

Thermal expansion is within acceptable limits and does not compromise structural integrity.

Connecting Rod – Fatigue Durability

- Predicted Fatigue Life: **1×10^{9} cycles**
- Minimum Fatigue Safety Factor: **6.36**

The connecting rod operates in the infinite life region, indicating robust durability under cyclic loading.

Crankshaft – Dynamic Stability

- First Natural Frequency: **323.65 Hz**
- Operating Frequency: **41.67 Hz**
- Frequency Separation Factor: **7.8×**

Operating speed lies well below resonance region, ensuring stable vibration performance.

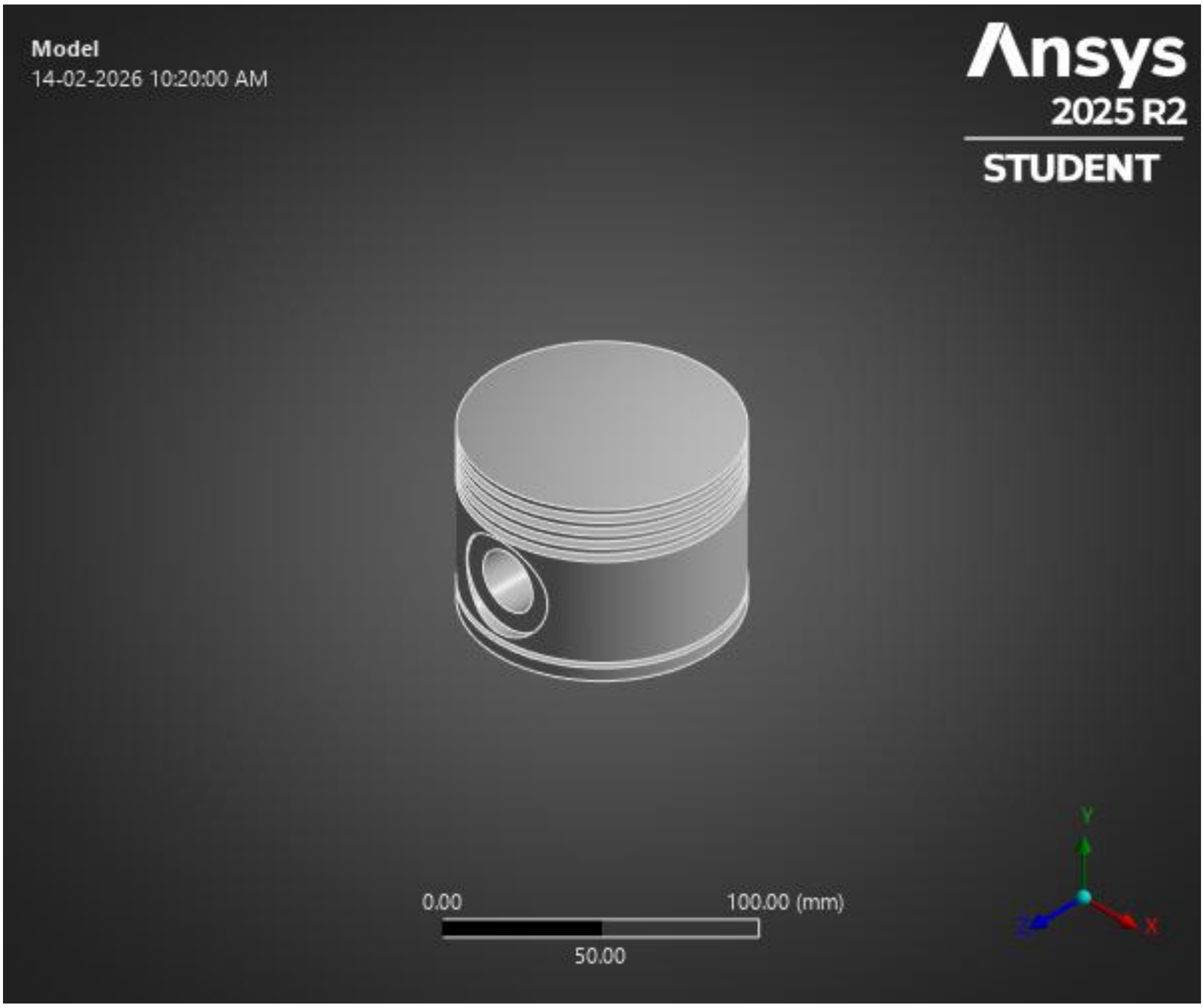
Overall Conclusion

The radial engine assembly demonstrates adequate structural strength, thermal stability, fatigue durability, and dynamic safety under assumed peak combustion and operating conditions.

All evaluated components operate within acceptable engineering limits, with sufficient safety margins against yielding, fatigue failure, and resonance.

The design is considered suitable for safe operation under the defined assumptions.

Structural – Piston



Piston – Load Justification

Engine Assumptions

Parameter	Value	Justification
Peak Combustion Pressure	6 MPa	Typical SI aircraft engine peak
Engine Speed	2500 RPM	Cruise operating condition
Piston Material	Aluminium Alloy	
Yield Strength	280 MPa	

Combustion Force Calculation

$$F = P \times A$$

Assuming bore ≈ 90 mm:

$$A = \frac{\pi D^2}{4}$$
$$A = 0.00636\, m^2$$
$$F = 6 \times 10^6 \times 0.00636$$
$$F \approx 38{,}160\, N \approx 38\, kN$$

The piston experiences approximately 38 kN compressive load at peak combustion.
This load was applied as uniform pressure on the piston crown in ANSYS to simulate peak firing condition.

Mesh Information

Mesh Summary

Element Type	Tetrahedral (ANSYS default)
Nodes	~33,335
Solver	Static Structural
Load Type	Uniform Pressure on crown
Constraints	Pin-bore constrained

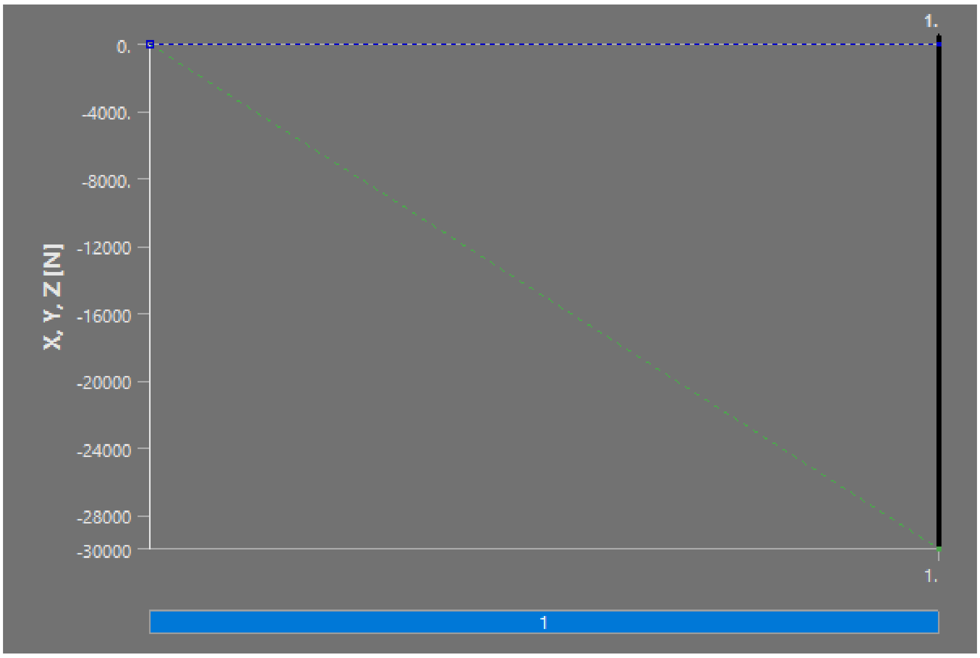
Quality Note

Mesh density is sufficient for global stress evaluation.
Future refinement recommended near:

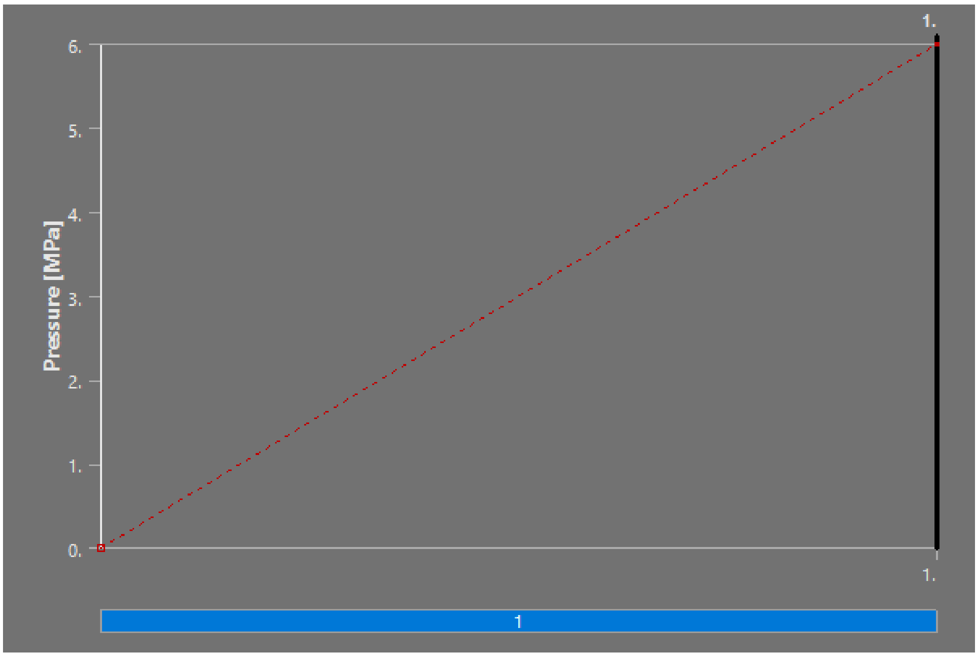
- Pin boss fillet
- Crown-to-skirt transition

Input Boundary Conditions

Applied Axial Combustion Force Boundary Condition

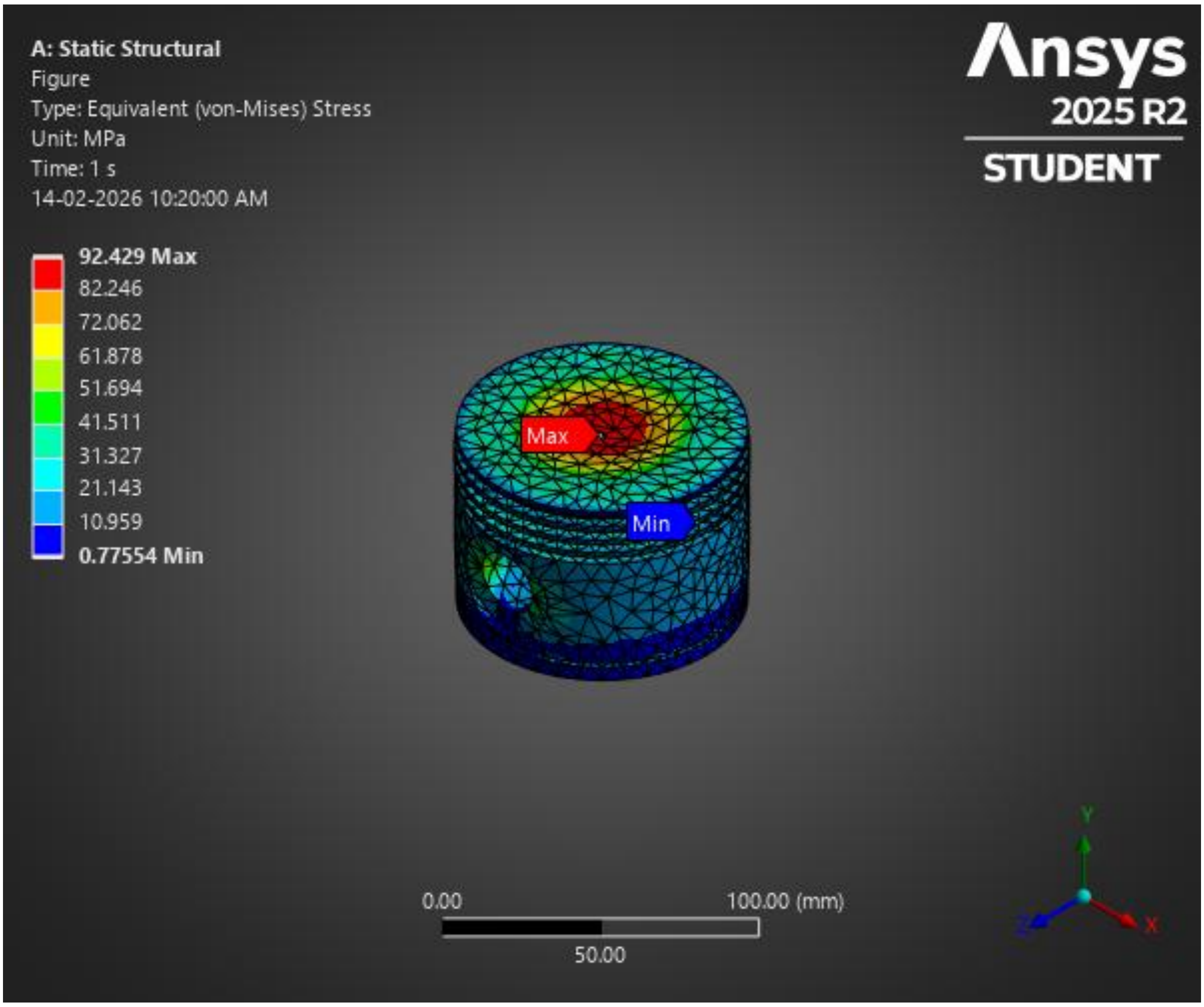


Applied Combustion Pressure Boundary Condition



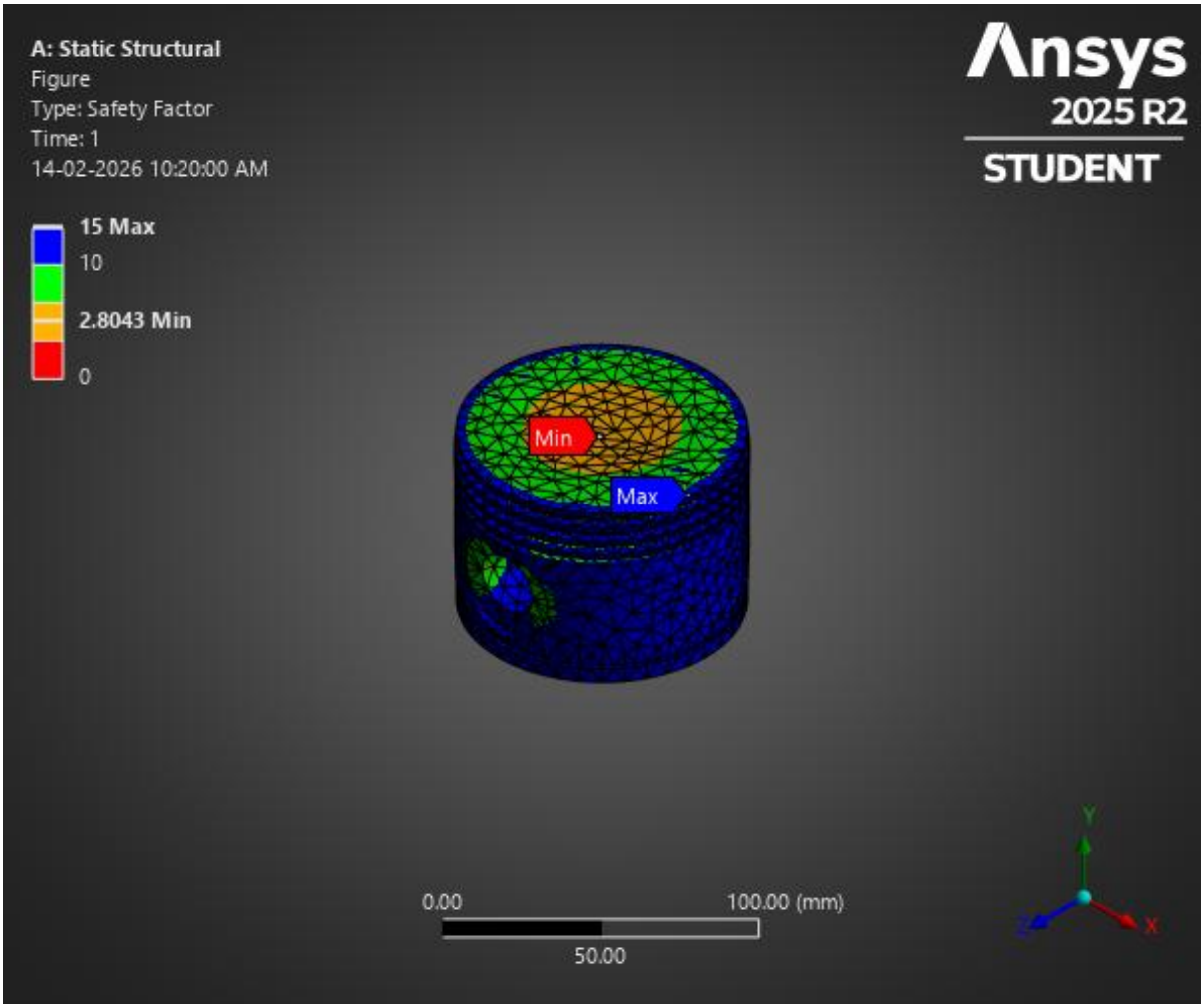
Results

Piston Deformation Under Peak Combustion Load



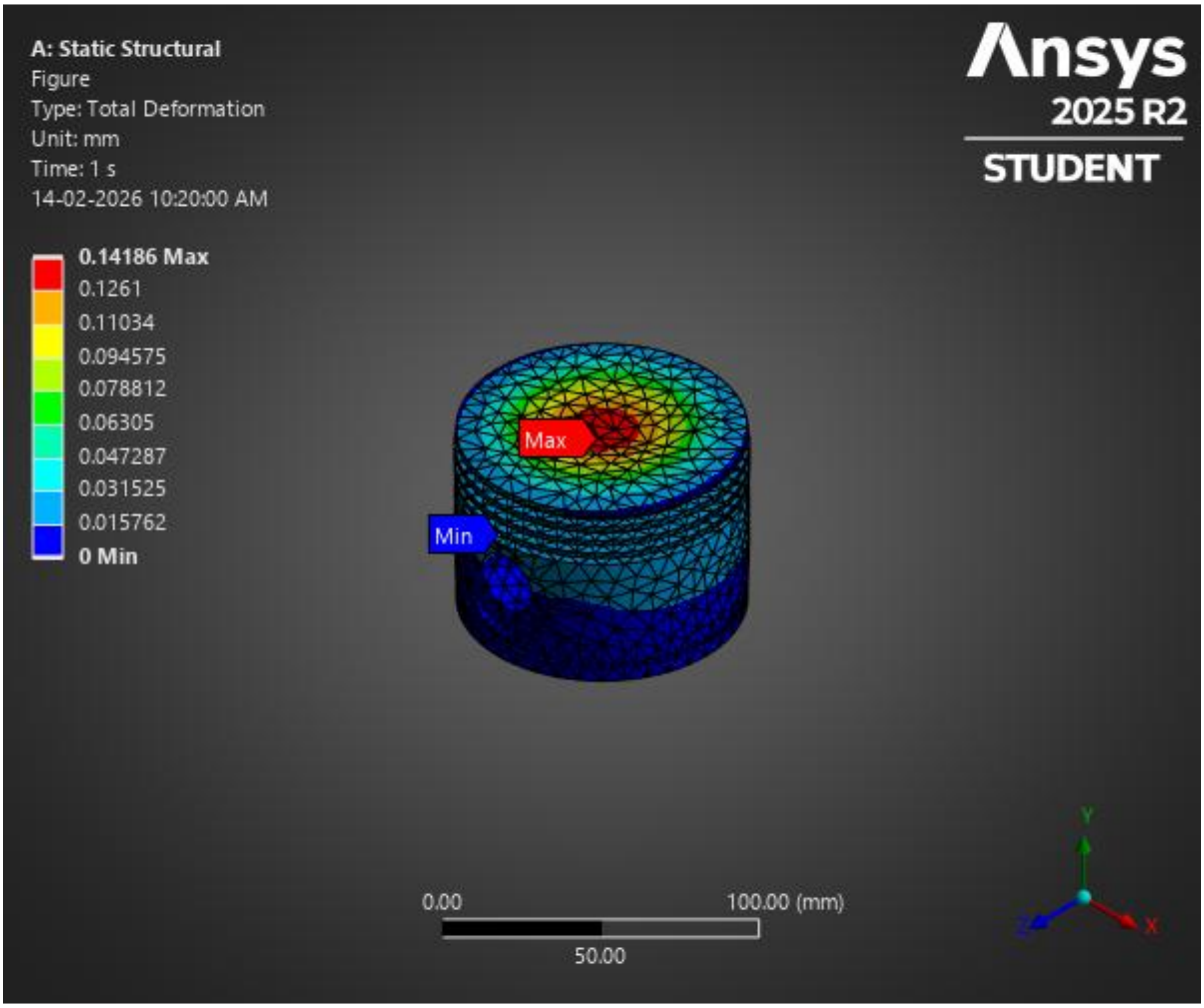
Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	0.	0.14186	2.1272e-002

Von Mises Stress Distribution Under Peak Combustion Load



Time [s]	Minimum [MPa]	Maximum [MPa]	Average [MPa]
1.	0.77554	92.429	25.804

Factor of Safety Assessment – Peak Combustion Load



Time [s]	Minimum	Maximum	Average
1.	2.8043	15.	10.781

Structural Results – Key Values

Result	Value
Maximum Von Mises Stress	92.43 MPa
Yield Strength	280 MPa
Minimum Safety Factor	2.8
Maximum Deformation	0.141 mm

Engineering Interpretation – Piston

Stress Behaviour

- Maximum stress observed near pin-boss fillet region.
- Stress concentration expected due to geometry transition.
- Peak stress = 92 MPa, which is:

$$\frac{92}{280} \approx 33\%$$

Only **33%** of yield strength.

Deformation

- Maximum deformation = **0.141 mm**
- Deformation is small relative to piston dimensions.
- No excessive crown distortion observed.

Safety Evaluation

- Minimum Safety Factor = **2.8**
- Design considered structurally safe under peak combustion loading.

Modal Relevance Check

Although piston itself was not modal-tested independently, system RPM comparison is important.

Engine speed:

$$2500\ RPM = 41.67\ Hz$$

Crankshaft first natural frequency: **323.65 Hz**

$$\frac{323.65}{41.67} \approx 7.8$$

Operating frequency is nearly 8× lower than first natural frequency.

Conclusion

No resonance risk under normal operating condition.

Summary

Key Highlights

Peak Load Applied	38 kN
Max Stress	92 MPa
Yield Strength	280 MPa
Safety Factor	2.8
Max Deformation	0.141 mm

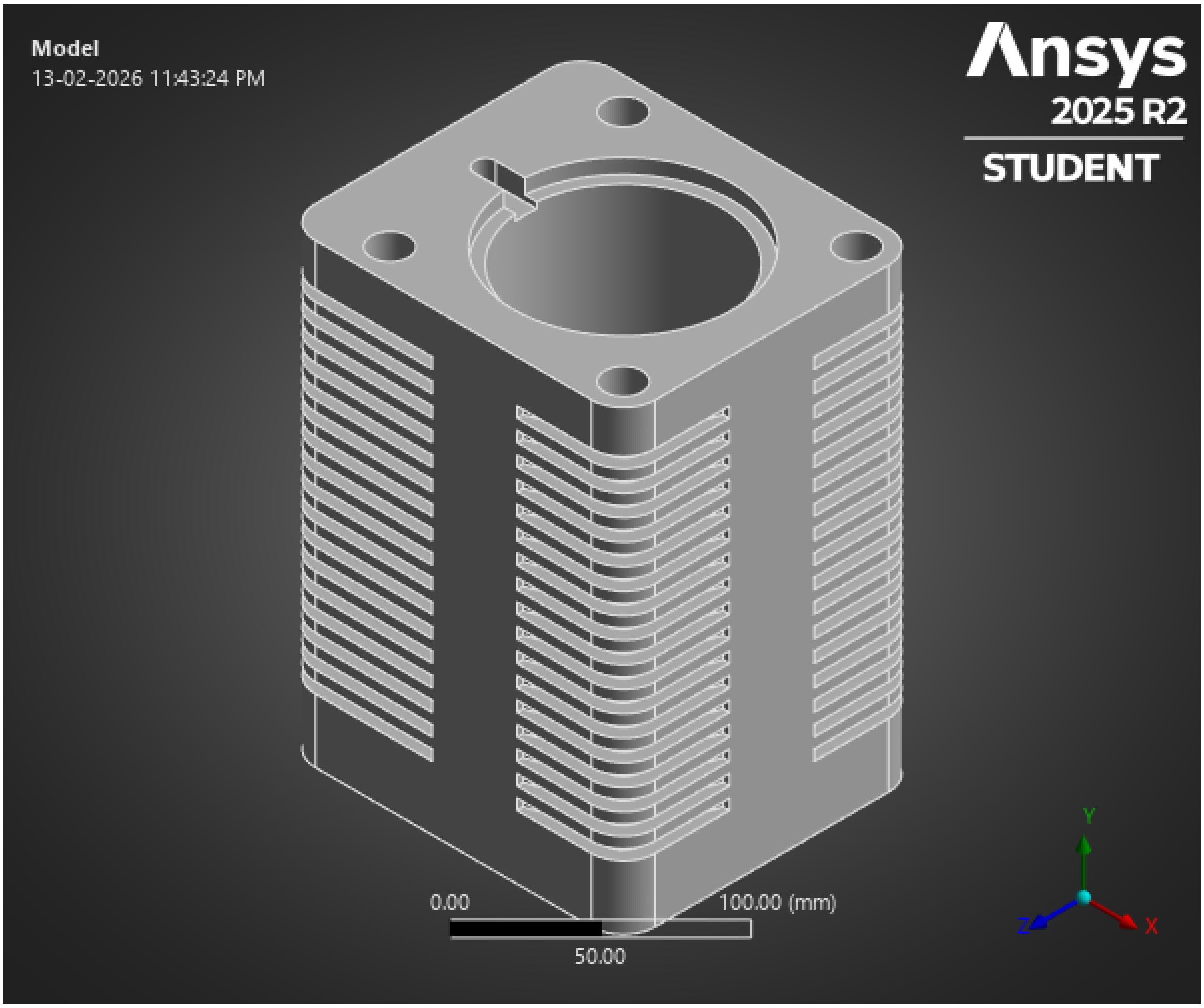
No resonance risk in system

Final Engineering Conclusion

The piston design demonstrates adequate structural integrity under assumed peak combustion pressure of **6 MPa**.

Stress levels remain well below yield limit, deformation is minimal, and overall safety margin is acceptable for operational conditions.

Thermal – Cylinder



Cylinder – Load Justification (Thermal)

Operating Assumptions

Parameter	Value	Justification
Ambient Temperature	21°C	Standard atmospheric condition
Convection Applied	5×10^{-6} W/mm^2·°C	Natural / air-cooled assumption
Material	Aluminium Alloy	
Analysis Type	Steady-State Thermal	

During combustion:

- Heat transfers from combustion chamber to cylinder wall.
- Outer surface dissipates heat via convection (air cooling).
- Thermal gradients induce expansion and thermal strain.

Thermal results were imported into structural analysis to evaluate thermally induced stresses and deformation.

Mesh Information

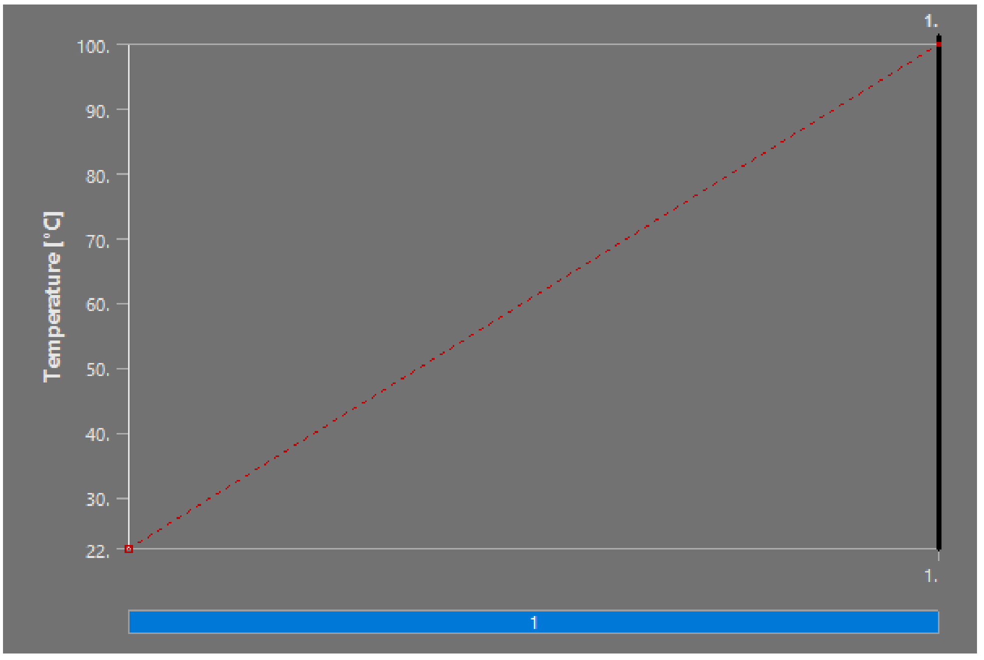
Mesh Summary

Element Type	Tetrahedral
Node Count	~33,335 (100% mapped during load transfer)
Solver	Steady-State Thermal → Static Structural (Coupled)
Load Transfer Mapping	100% successful
Centroid Distance Error	0.0099% (negligible)

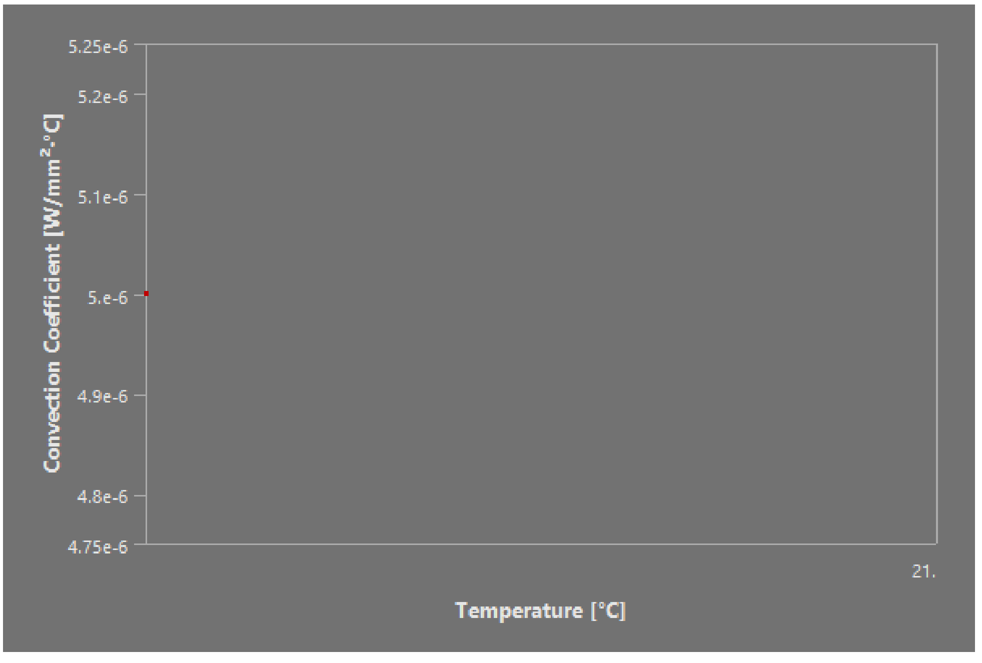
Load transfer accuracy is excellent.
No interpolation loss observed between thermal and structural domains.

Input Boundary Conditions

Applied Thermal Boundary Condition – Combustion-Side Temperature



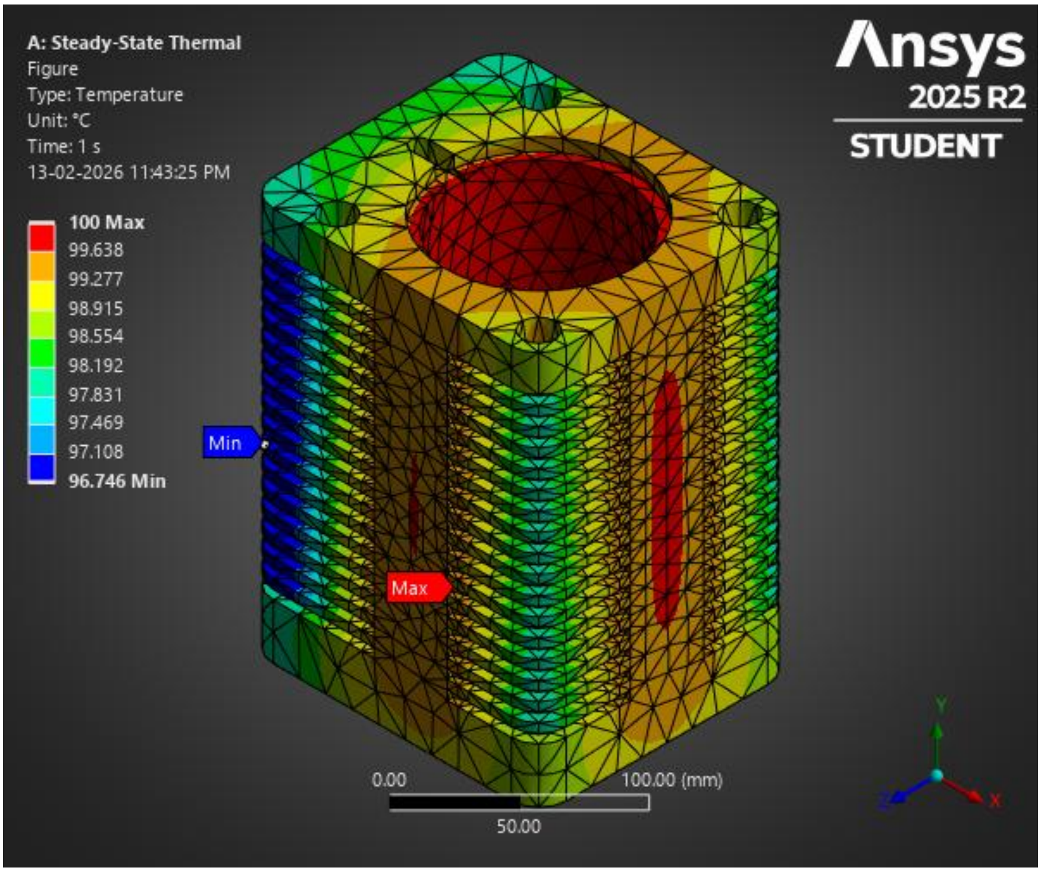
Applied Convective Heat Transfer Boundary Condition



Temperature [°C]	Convection Coefficient [W/mm²·°C]
21.	5.e-006

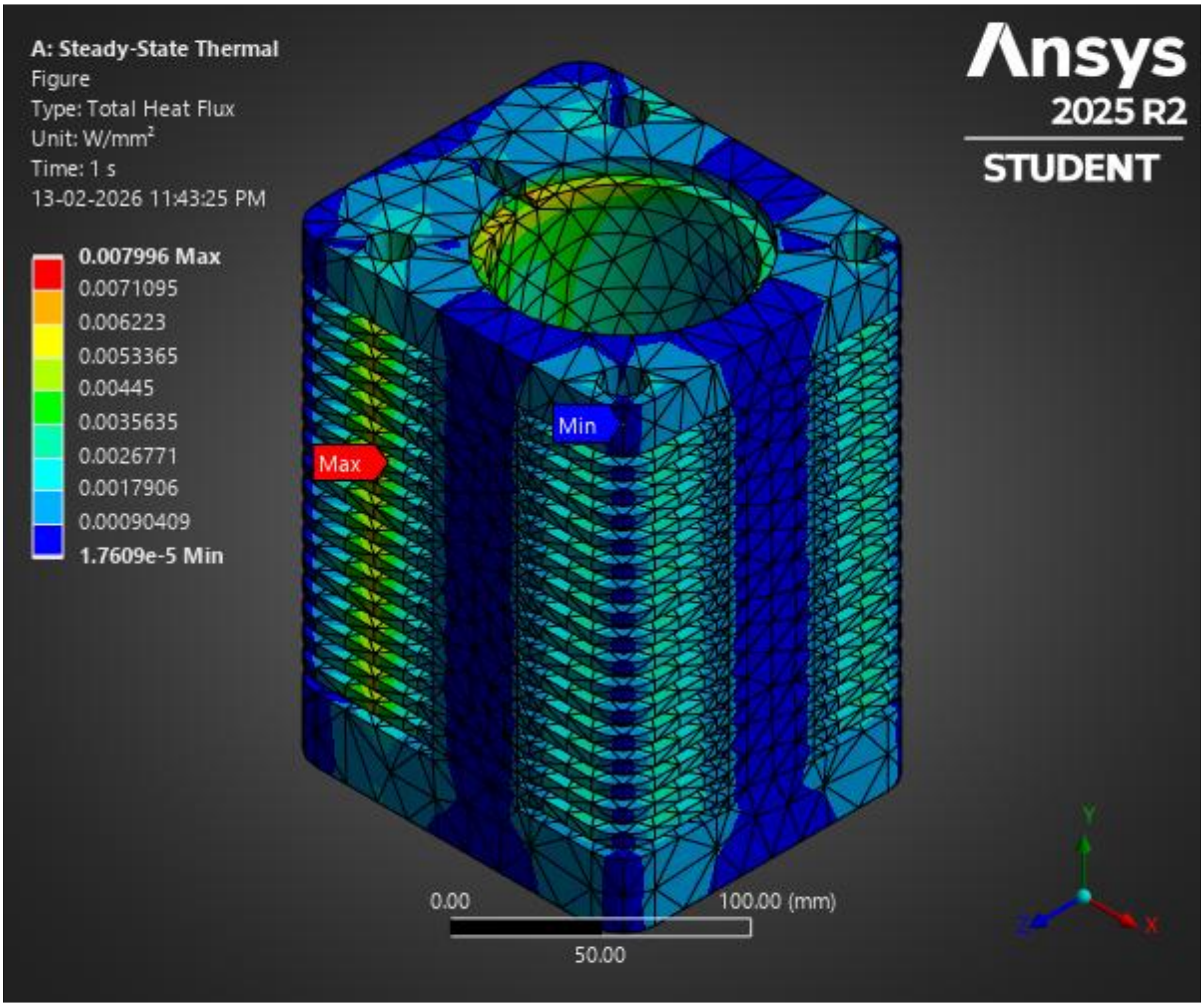
Results

Temperature Field Under Steady-State Thermal Loading



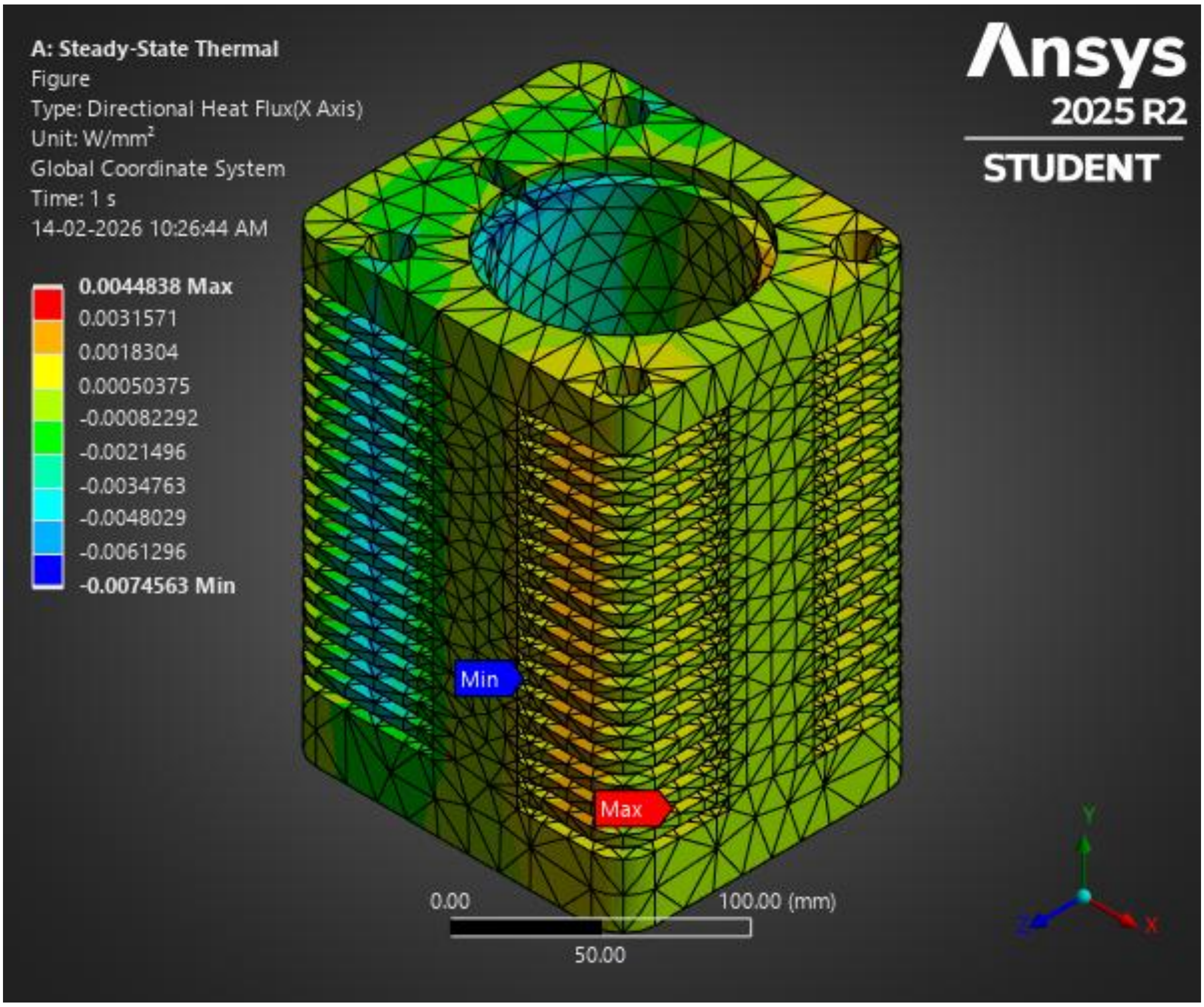
Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1.	1.7609e-005	7.996e-003	2.0703e-003

Total Heat Flux Distribution Under Steady-State Thermal Loading



Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1.	1.7609e-005	7.996e-003	2.0703e-003

Directional Heat Flux Distribution Under Steady-State Thermal Loading



Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1.	-7.4563e-003	4.4838e-003	-4.6801e-004

Imported Inputs - Thermal Results

Cylinder - Static Structural - Imported Load - Imported Body Temperature - Imported Load Transfer Summary

Aluminium Alloy - Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

Aluminium Alloy - Isotropic Thermal Conductivity

Thermal Conductivity W mm^-1 K^-1	Temperature K
0.114	173.15
0.144	273.15
0.165	373.15
0.175	473.15

Thermal Results – Key Values

Result	Value
Maximum Thermal Deformation	0.2386 mm
Average Thermal Deformation	0.145 mm
Thermal Strain	1.77×10^{-3} mm/mm
Convection Coefficient	5×10^{-6} W/mm^2·°C

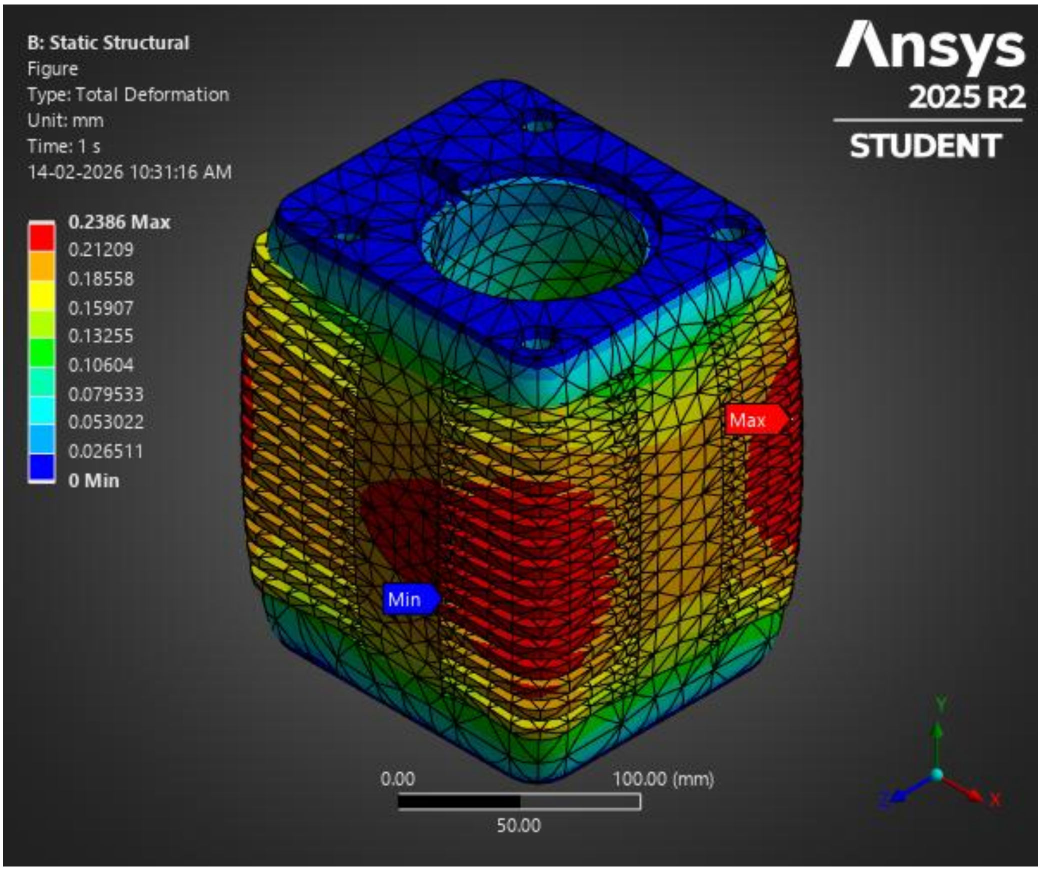
Engineering Interpretation – Cylinder

Thermal Behaviour

- Thermal expansion is consistent with Aluminium CTE.
- Maximum deformation = **0.2386 mm.**
- Expansion appears uniformly distributed across cylinder geometry.
- No severe localized temperature gradients observed.

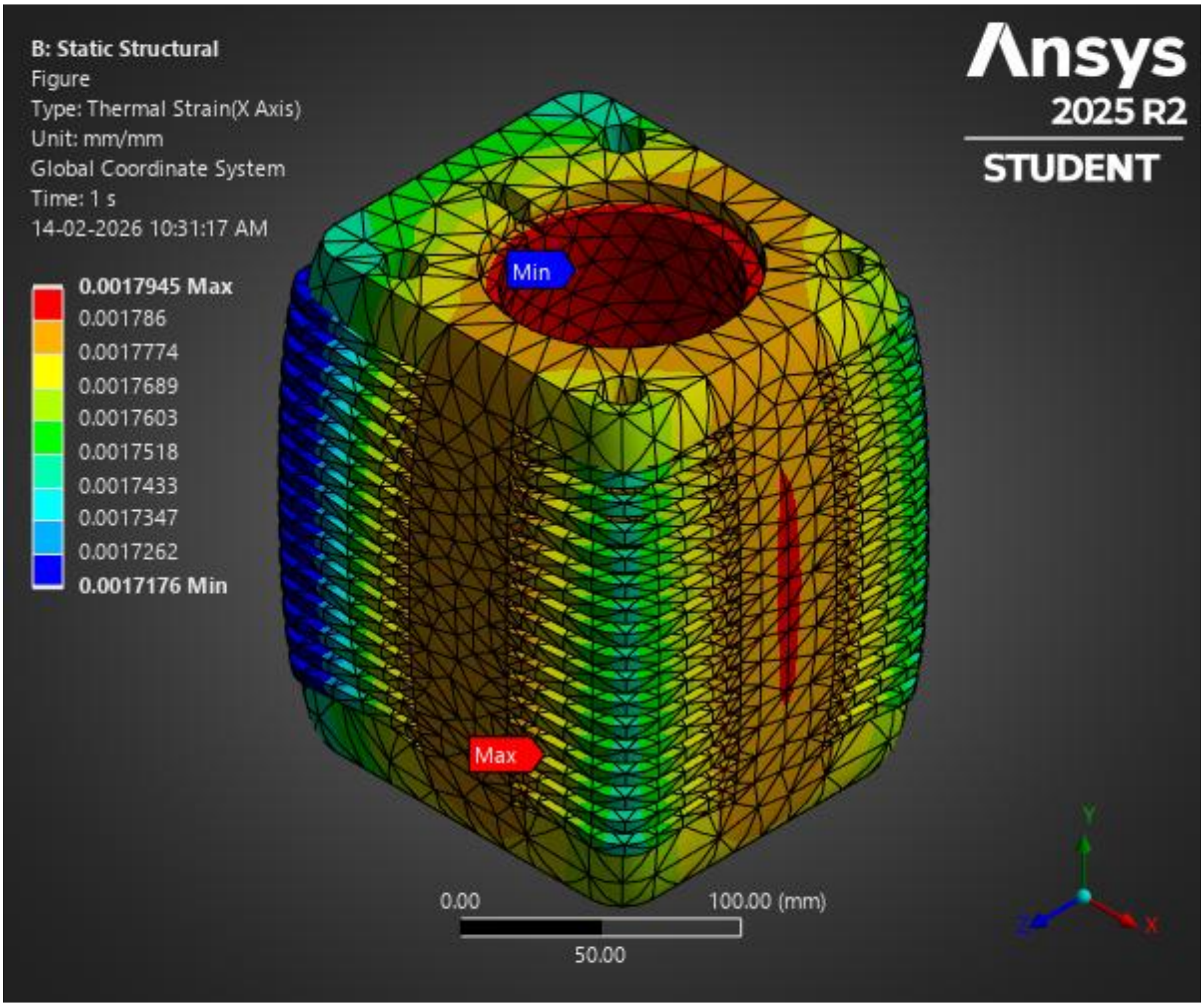
Results

Total Deformation Under Thermally Induced Loading



Time [s]	Minimum [mm]	Maximum [mm]	Average [mm]
1.	0.	0.2386	0.14518

Thermal Strain Under Thermally Induced Loading



Time [s]	Minimum [mm/mm]	Maximum [mm/mm]	Average [mm/mm]
1.	1.7176e-003	1.7945e-003	1.7717e-003

Engineering Safety Assessment

- Thermal deformation < 0.25 mm.
- No indication of excessive stress amplification.
- Material yield (280 MPa) not exceeded.

The cylinder design is thermally stable under assumed Steady-State conditions.

Modal Relevance Check

Engine operating frequency:

$$\mathbf{2500\ RPM = 41.67\ Hz}$$

Crankshaft first natural frequency: **323.65 Hz.**

Operating speed is significantly below system resonance region.

Thermal expansion does not significantly influence dynamic stability.

Summary

Key Highlights

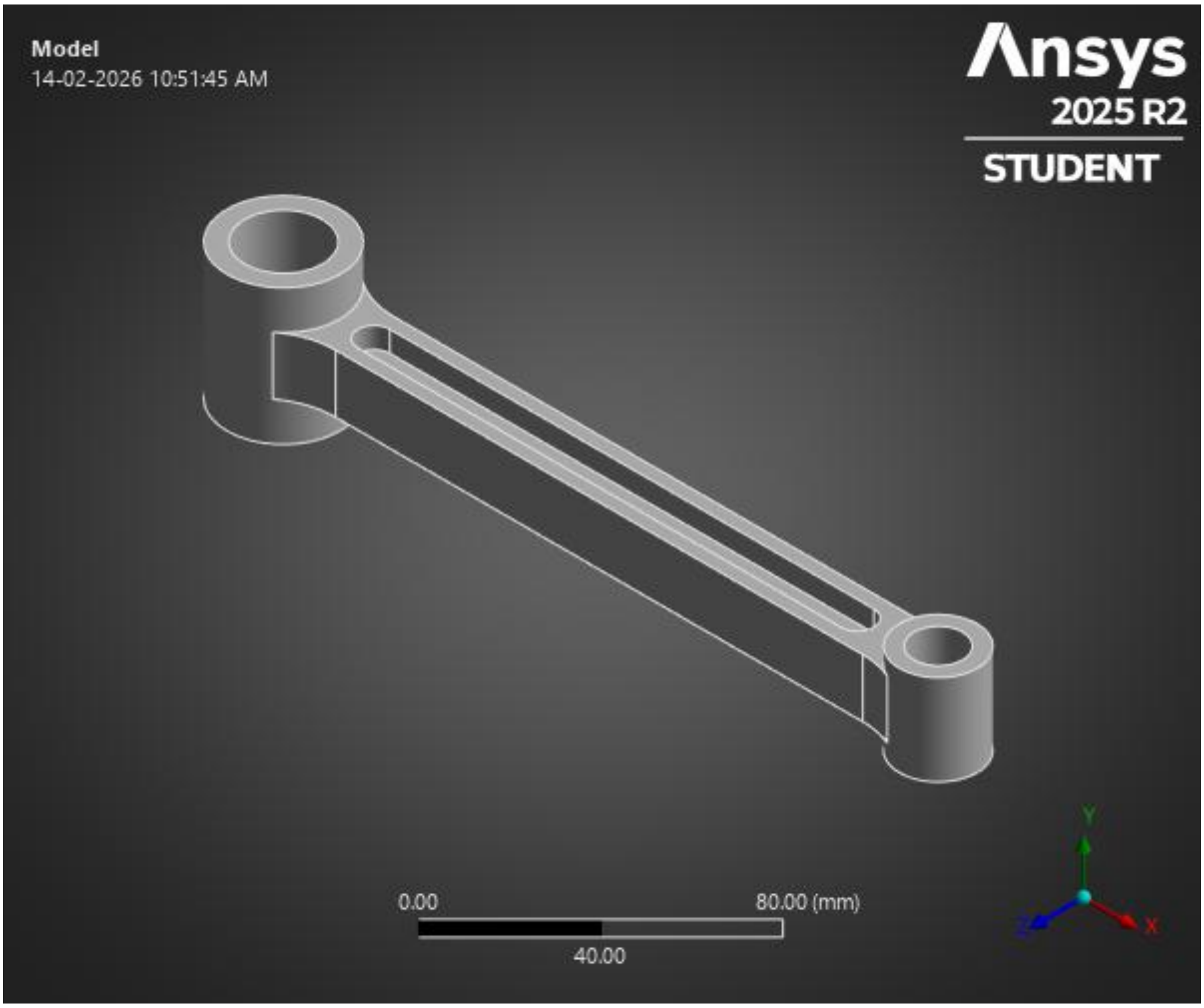
- Thermal deformation: 0.2386 mm
- Thermal strain: 1.77×10^{-3}
- No excessive distortion
- Stable under operating RPM

Final Engineering Conclusion

The cylinder demonstrates stable thermal behavior under Steady-State operating assumptions.

Thermal expansion is within acceptable range and does not compromise structural integrity or dynamic stability of the engine assembly.

Fatigue – Connecting Rod



Connecting Rod – Load Justification

Operating Assumptions

Parameter	Value	Justification
Peak Combustion Pressure	6 MPa	Typical SI aircraft engine peak
Engine Speed	2500 RPM	Cruise operating condition
Equivalent Piston Force	≈ 38 kN	
Material	Structural Steel	
Analysis Type	Static + Fatigue	

Combustion force generated at piston:

$$\boldsymbol{F = P \times A \approx 38\ kN}$$

This load is transmitted axially through the connecting rod to the crankshaft.

During operation, the rod experiences:

- Compressive load (power stroke)
- Tensile load (inertia during return stroke)
- Cyclic loading at engine frequency

At 2500 RPM:

$$\frac{\mathbf{2500}}{\mathbf{60}} = \mathbf{41.67}\ \textbf{cycles/sec}$$

So, fatigue evaluation is essential.

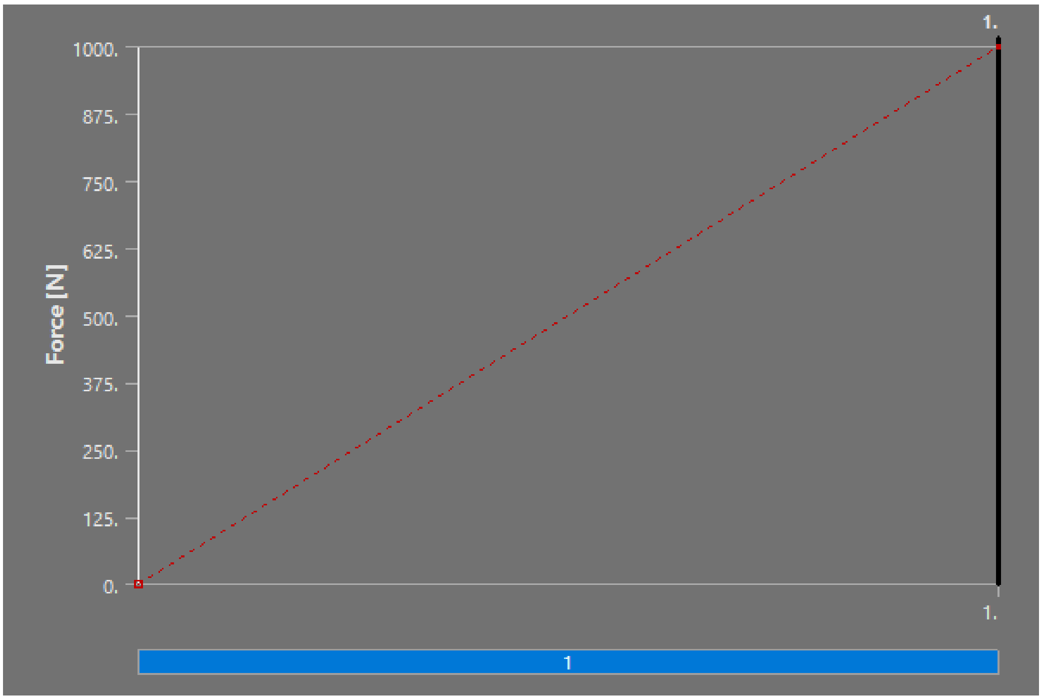
Mesh Information

Mesh Summary

Element Type	Tetrahedral
Solver	Static Structural → Fatigue Tool
Load Type	Axial force at small end
Constraint	Big end bore constrained

Input Boundary Conditions

Applied Combustion Force Boundary Condition

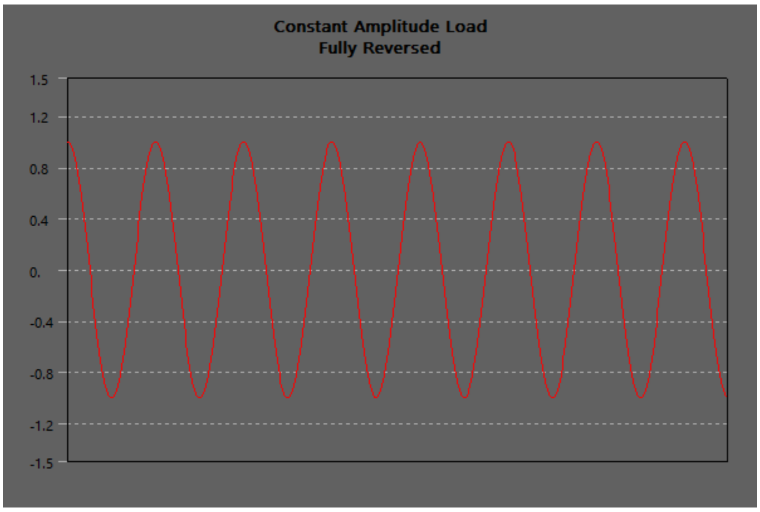


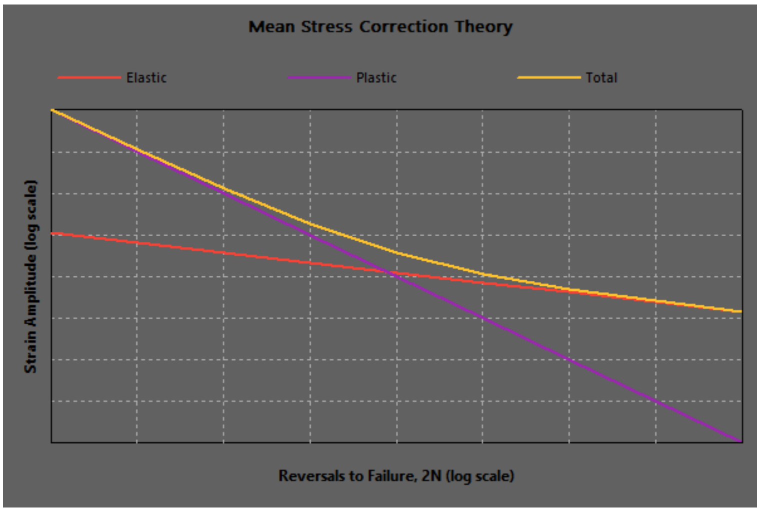
Static Structural Results – Key Values

(Used as input for fatigue)

Result	Value
Peak Stress	Within elastic range
Material Yield	280 MPa (reference)
Stress Concentration	Fillet region

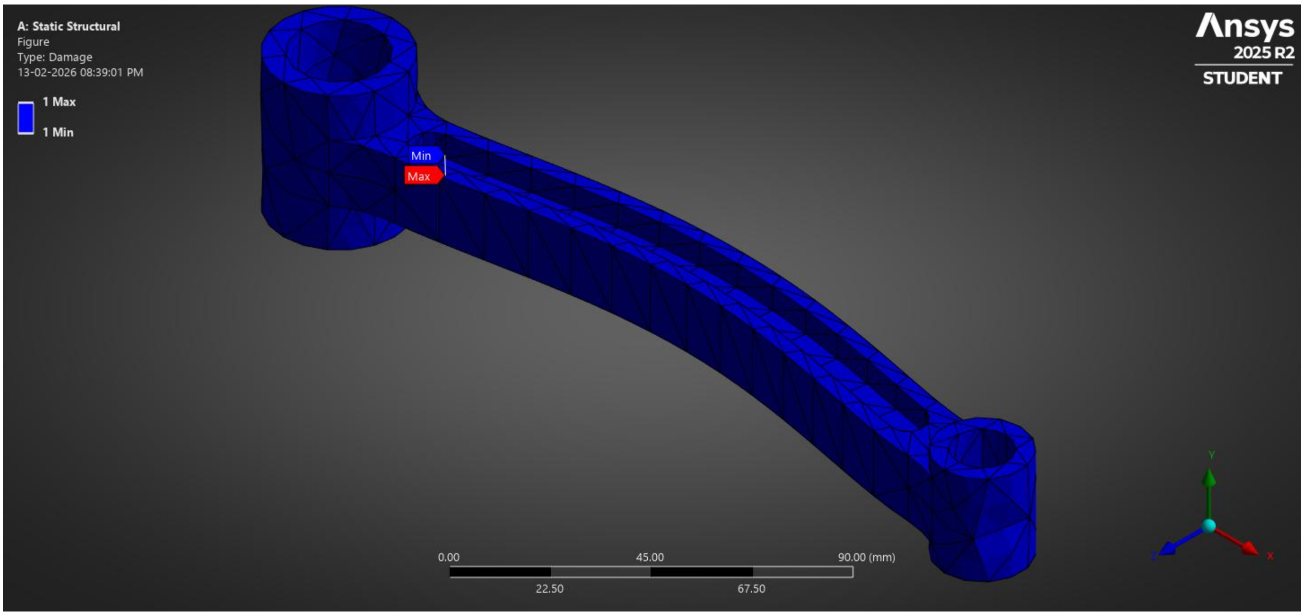
Fatigue Tool - Modal Natural Frequency



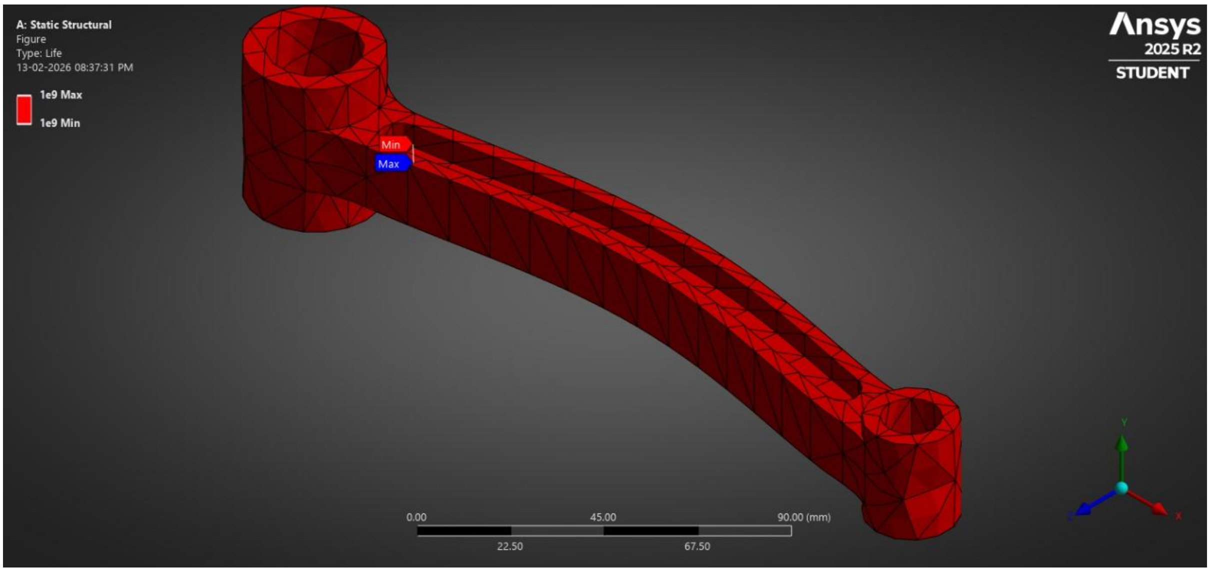


Results

Cumulative Fatigue Damage Assessment

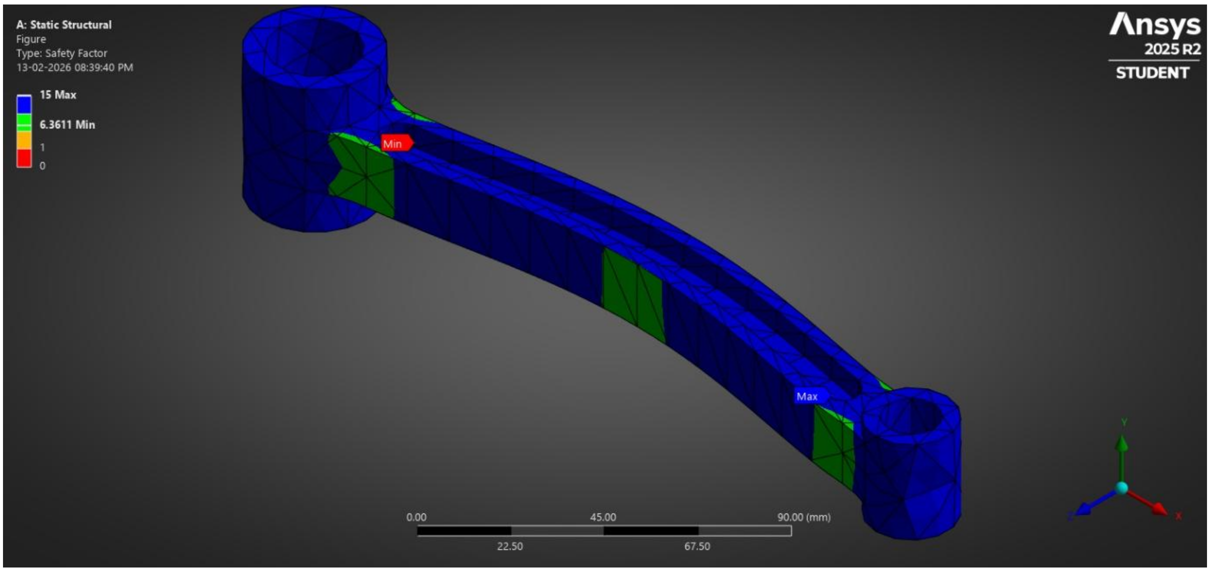


Fatigue Life Prediction Under Cyclic Combustion Loading



Time [s]	Minimum	Maximum	Average
1.	1.e+009	1.e+009	1.e+009

Fatigue Safety Margin Assessment



Time [s]	Minimum	Maximum	Average
1.	6.3611	15.	14.213

Structural Steel - S-N Curve

Alternating Stress MPa	Cycles
3999	10
2827	20
1896	50
1413	100
1069	200
441	2000
262	10000
214	20000
138	1.e+005
114	2.e+005
86.2	1.e+006

Structural Steel - Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

Fatigue Results – Key Values

Result	Value
Predicted Life	1×10^9 cycles
Minimum Safety Factor	6.36
Damage	Negligible

Material Data Used:

- S–N curve defined.
- Mean stress = **0 MPa.**
- Strain-life parameters included.

Engineering Interpretation – Fatigue

Fatigue Behaviour

- Predicted life = **10^9 cycles.**
- This lies in the infinite life region for structural steel.
- Minimum fatigue safety factor = **6.36**, indicating highly conservative design.

Stress Distribution

- Maximum stress occurs near big-end fillet region.
- Expected due to geometric stress concentration.

Operational

At 2500 RPM:

$$\mathbf{41.67\ cycles/sec}$$
$$\mathbf{Cycles\ per\ hour = 150,000}$$

To reach 10^9 cycles:

$$\mathbf{\approx 6667\ hours\ of\ operation}$$

Which is far beyond typical engine service interval.

Durability Assessment

The connecting rod demonstrates:

- High fatigue resistance
- Safe stress levels
- Large safety margin

Design is over-safe and may allow weight optimization in future iterations.

Summary

Key Highlights

Applied Load	≈ 38 kN
Fatigue Life	1×10^9 cycles
Safety Factor	6.36

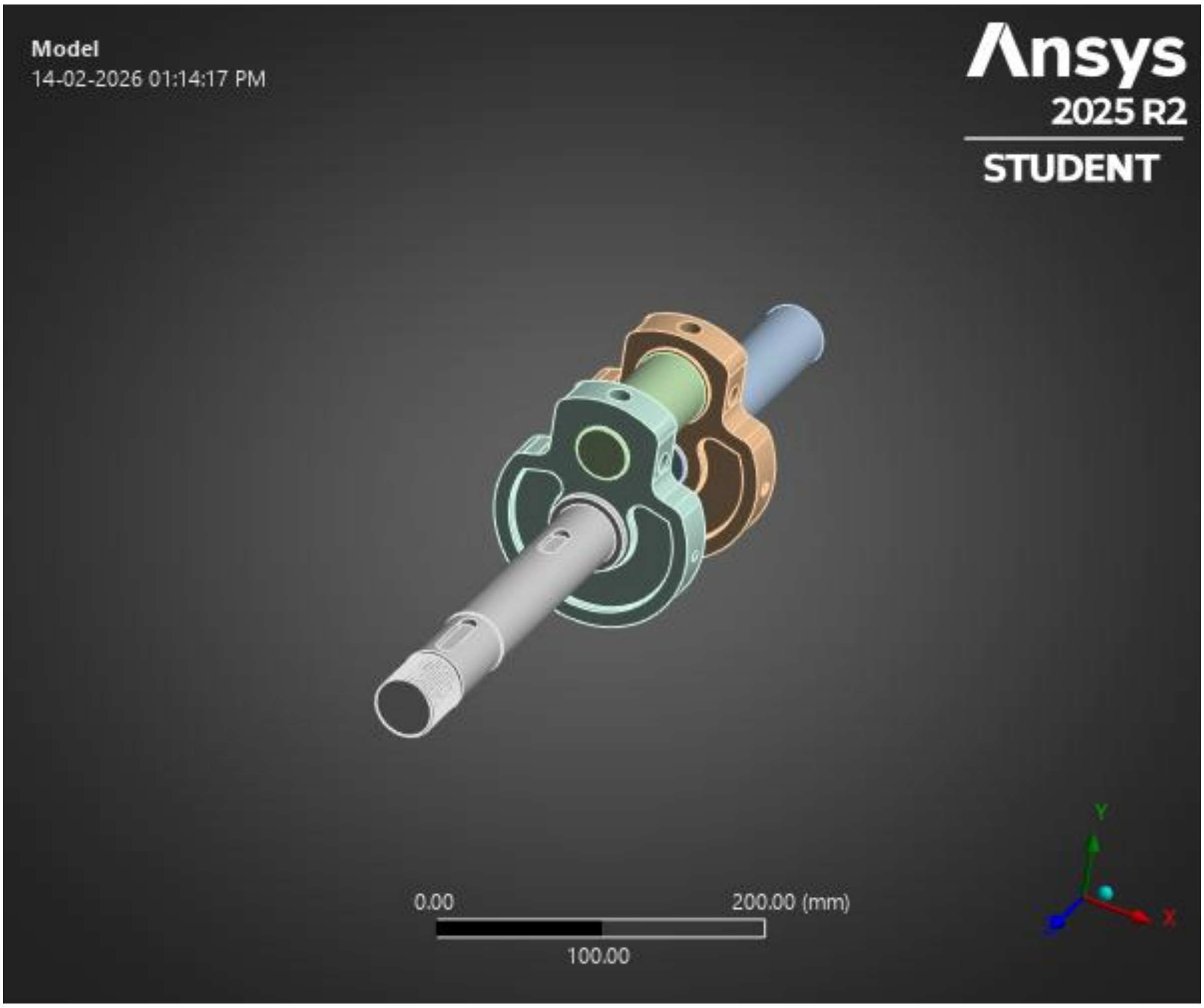
No critical fatigue damage predicted

Final Engineering Conclusion

The connecting rod is structurally and fatigue-safe under peak combustion loading conditions at **2500 RPM**.

The predicted fatigue life significantly exceeds typical engine operational lifespan, indicating robust durability.

Modal – Crankshaft



Crankshaft – Load Justification

Operating Assumptions

Parameter	Value	Justification
Engine Speed	2500 RPM	Cruise condition
Operating Frequency	41.67 Hz	2500 / 60
Material	Structural Steel	
Analysis Type	Modal (Free Vibration)	

Analysis Justification

The crankshaft:

- Rotates continuously
- Experiences cyclic torque
- Is subjected to dynamic excitation from combustion events

If operating frequency approaches natural frequency → resonance may occur.

Resonance leads to:

- Excessive vibration
- Bearing wear
- Fatigue failure
- Catastrophic shaft fracture

Therefore, modal analysis is essential.

Mesh Information

Mesh Summary

Element Type	Tetrahedral
Solver	Modal (Eigenvalue Extraction)
Number of Modes Extracted	10
Constraint	Bearing journal supports applied

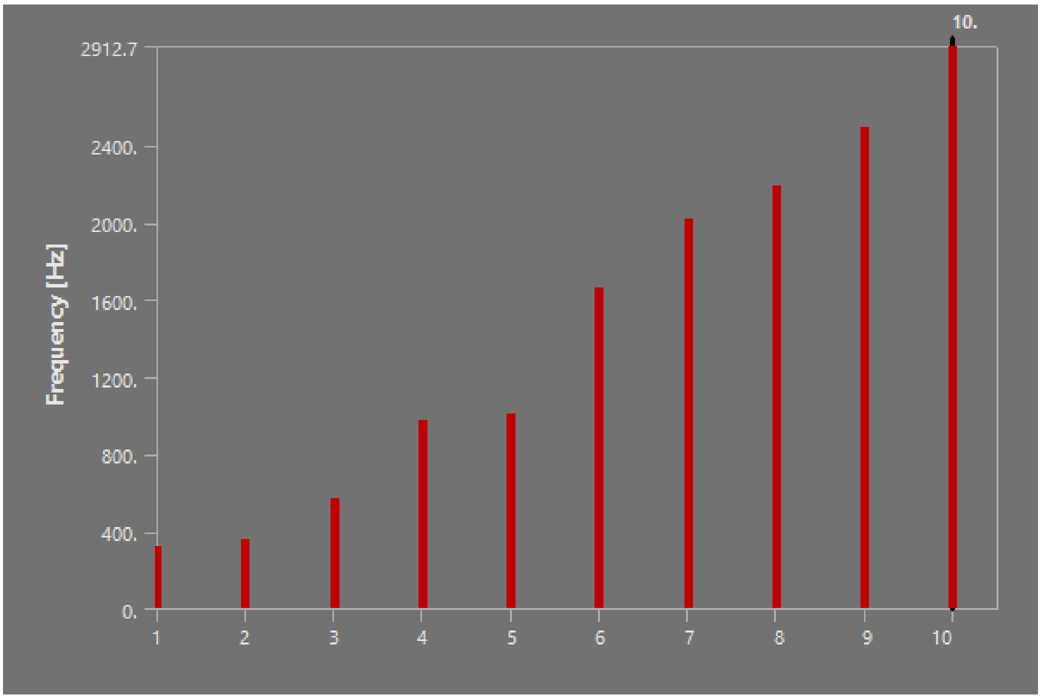
Critical Mesh Regions

- Fillets near crank webs
- Journal transitions
- Counterweight regions

Mesh density adequate for global frequency estimation.

Results

Natural Frequency Assessment for Resonance Avoidance



Mode	Frequency [Hz]
1.	323.65
2.	360.89
3.	568.44
4.	971.36
5.	1005.5
6.	1666.3
7.	2022.6
8.	2185.6
9.	2494.4
10.	2912.7

Operating Frequency Comparison

Engine speed

$$\mathbf{2500}\ \boldsymbol{RPM}$$

$$\frac{\mathbf{2500}}{\mathbf{60}} = \mathbf{41.67}\ \boldsymbol{Hz}$$

Resonance Check

$$\frac{\mathbf{323.65}}{\mathbf{41.67}} \approx \mathbf{7.8}$$

First natural frequency is nearly 8 times higher than operating frequency.

Engineering Interpretation – Modal

Dynamic Stability Assessment

- Operating frequency: **41.67 Hz**
- First natural frequency: **323.65 Hz**

Operating speed lies far below resonance region.

No frequency overlap observed within first 10 modes.

System-Level Observation

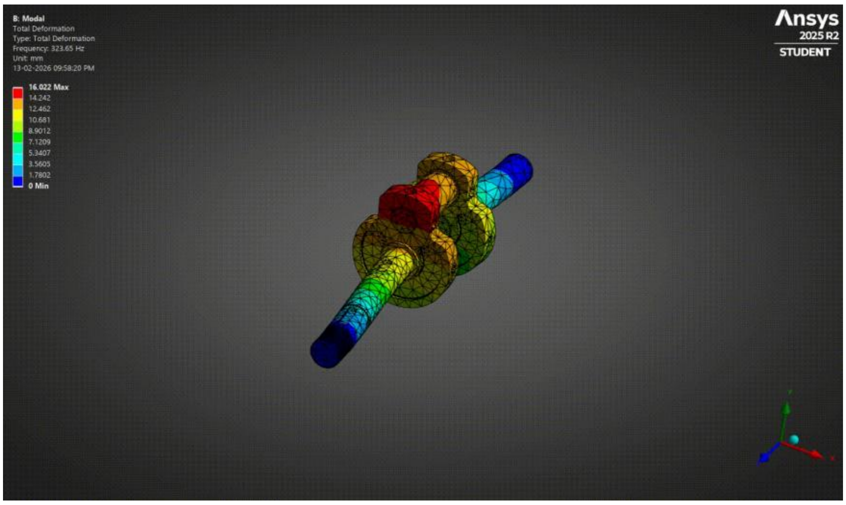
Even if RPM increases to: **4000 RPM**

$$\frac{\mathbf{4000}}{\mathbf{60}} = \mathbf{66.67}\ \boldsymbol{Hz}$$

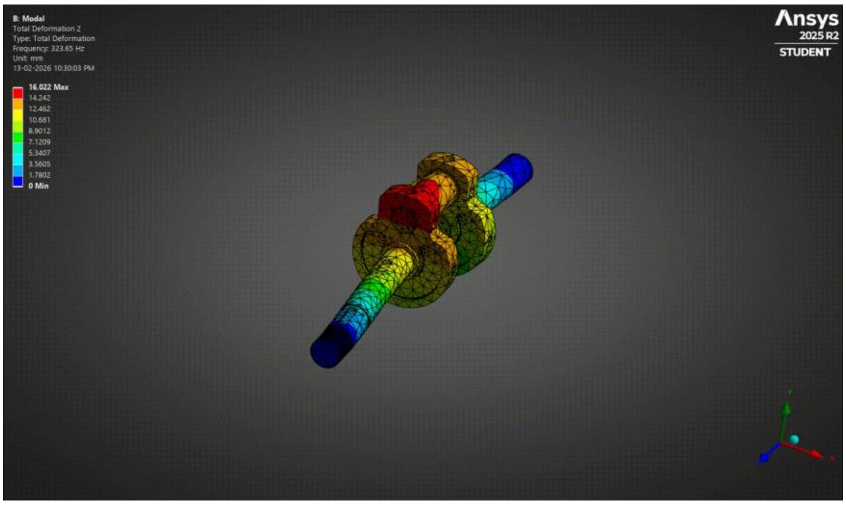
Still significantly below 323 Hz.

Thus, safe dynamic margin maintained.

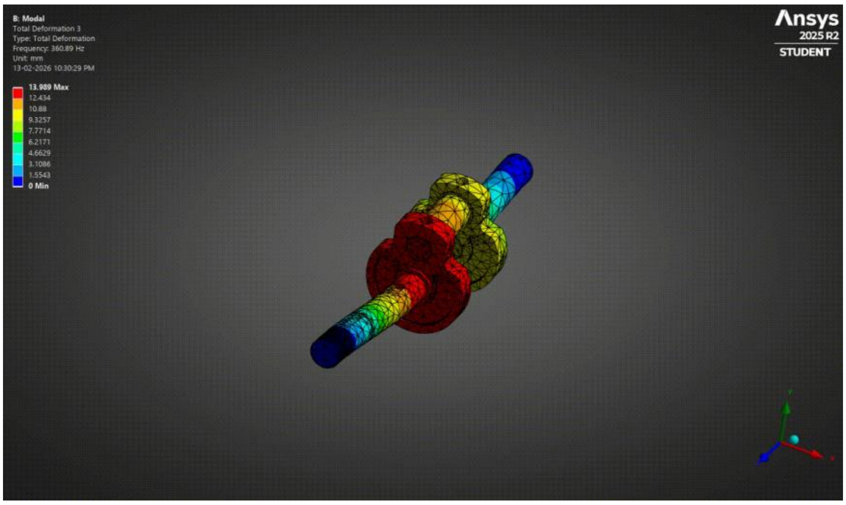
Frequency – 323.65 Hz



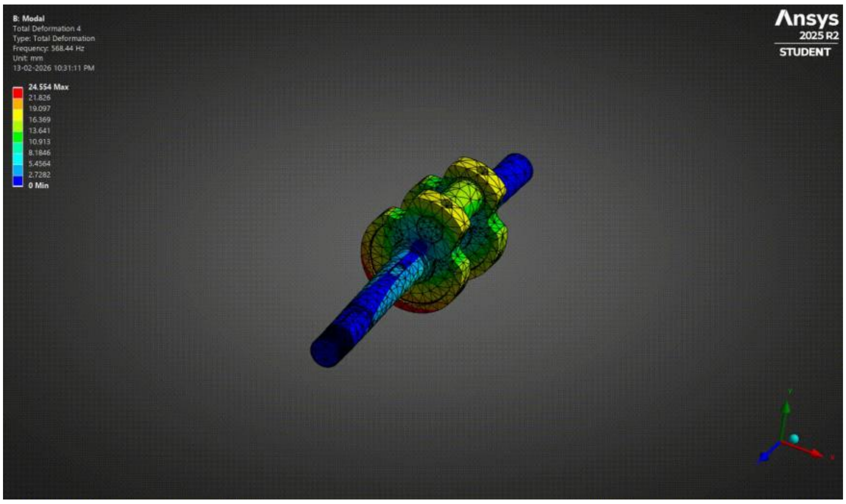
Frequency – 360.89 Hz



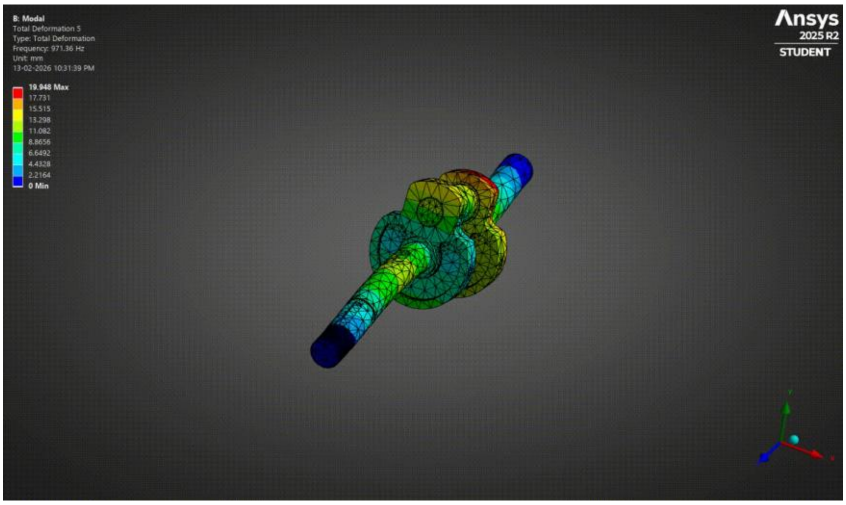
Frequency – 568.44 Hz



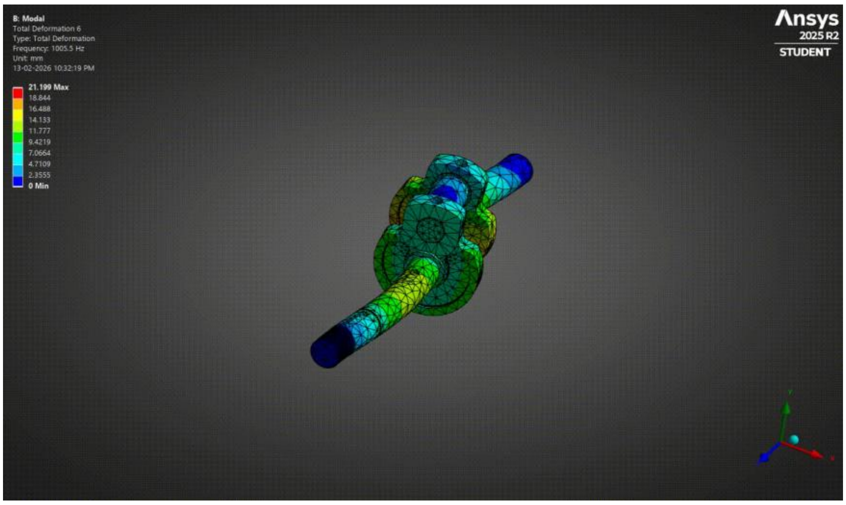
Frequency – 971.36 Hz



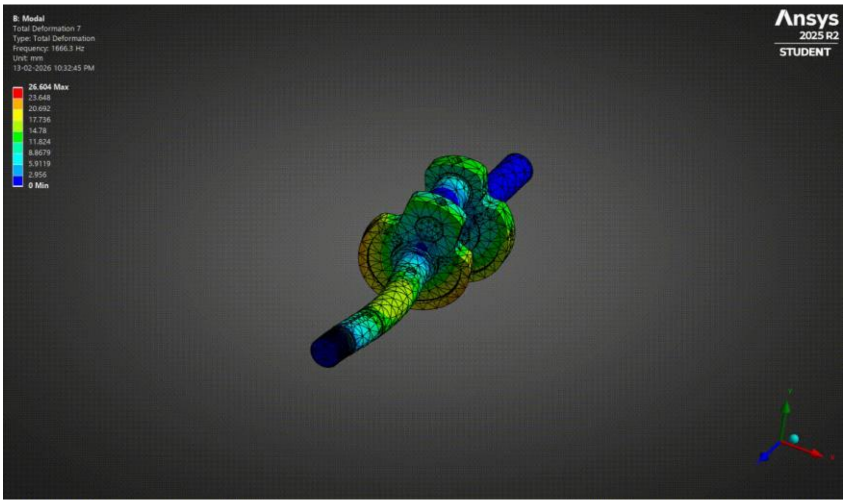
Frequency – 1005.5 Hz



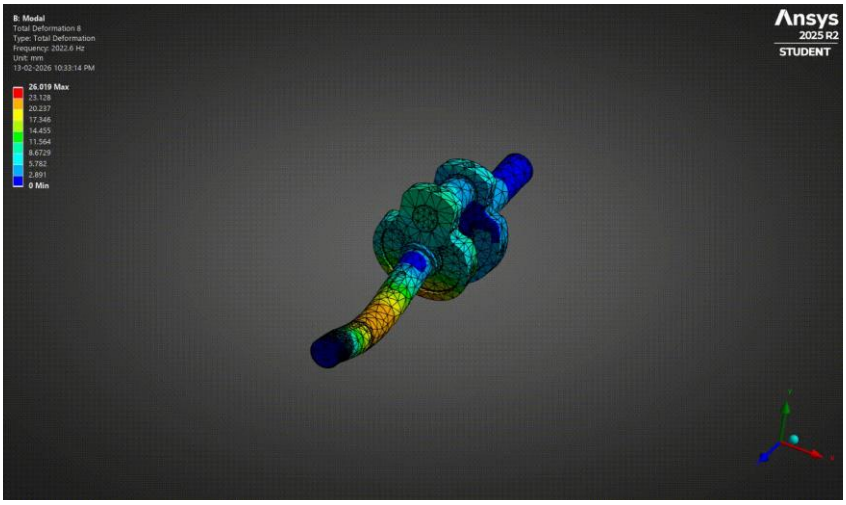
Frequency – 1666.3 Hz



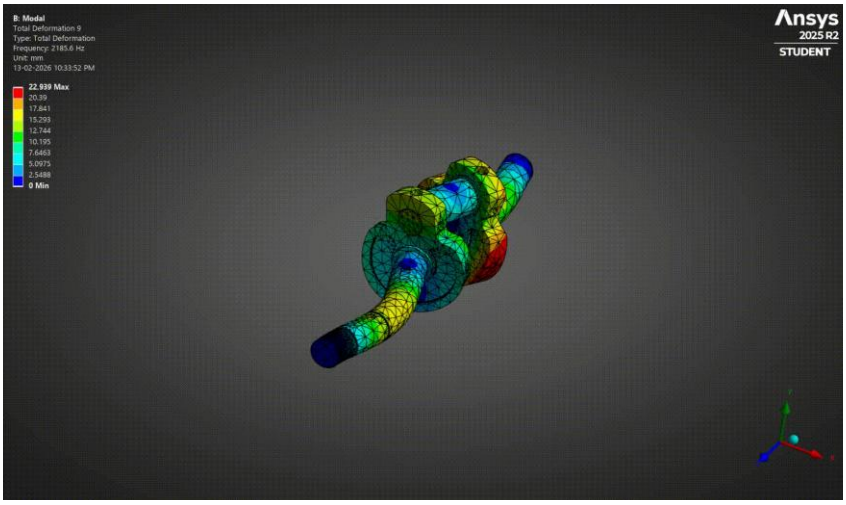
Frequency – 2022.6 Hz



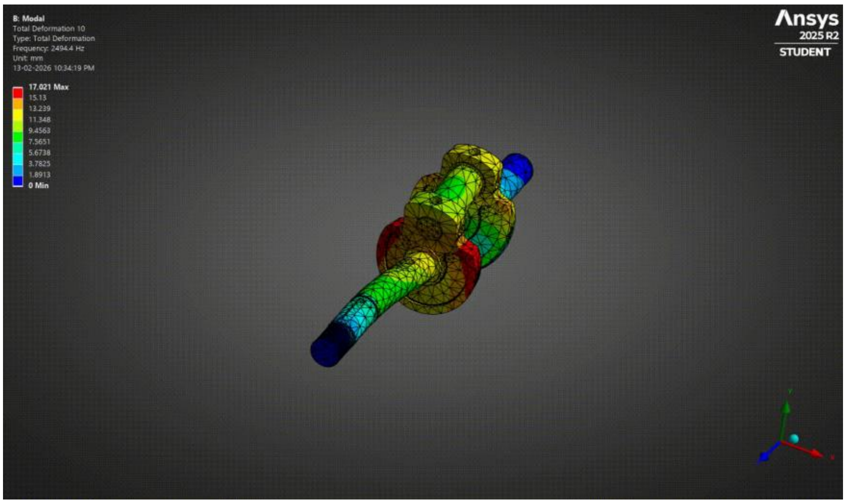
Frequency – 2185.6 Hz



Frequency – 2494.4 Hz



Frequency – 2912.7 Hz



Summary – Crankshaft Validation

Key Highlights

- Operating Frequency: **41.67 Hz**
- First Natural Frequency: **323.65 Hz**
- Separation Factor: **7.8×**
- No resonance risk detected.

Final Engineering Conclusion

The crankshaft exhibits stable dynamic characteristics under assumed operating conditions.

Natural frequencies are significantly higher than excitation frequency, ensuring safe vibration performance and minimal risk of resonance-induced fatigue.

Limitations of Study

Although the analysis provides valuable insight into structural, thermal, fatigue, and dynamic performance, the following simplifications were made:

Combustion Pressure Simplification

- Peak combustion pressure was assumed as a static uniform load (6 MPa).
- Real combustion pressure is:
  - Time-dependent
  - Non-uniform
  - Influenced by ignition timing and turbulence

Dynamic pressure fluctuations were not modeled.

No Transient Thermal Analysis

- A steady-state thermal condition was assumed.
- Real engines experience:
  - Transient heating during startup
  - Thermal cycling during operation
  - Cooling during shutdown

Thermal fatigue effects due to cyclic temperature variation were not evaluated.

No Contact Modelling

- Piston-cylinder contact interaction was not included.
- Bearing contact effects in crankshaft were simplified.
- No frictional heat generation was modeled.

Contact stress concentrations may differ under real operating conditions.

No Inertial & Rotational Effects

- Inertial forces due to reciprocating motion were not explicitly modeled.
- Crankshaft torsional harmonic excitation was not analyzed.
- Gyroscopic effects were not considered.

Only linear modal analysis was performed.

Linear Material Behavior Assumed

- Materials were assumed to behave elastically.
- Plastic deformation, creep, and strain-rate effects were not included.

Engineering Implication

The results represent a first-level design validation under simplified operating assumptions.

Further refinement using transient, nonlinear, and harmonic analyses would improve accuracy for production-level validation.

Design Improvement & Optimization Suggestions

Even though the design is safe, engineering optimization opportunities exist.

Piston Optimization

Observation:

- Maximum stress located near pin-boss fillet region.
- Safety factor = 2.8 (conservative).

Suggested Improvement:

- Increase fillet radius to reduce stress concentration.
- Alternatively, reduce crown thickness slightly to optimize weight.

Potential benefit:

- 5–10% weight reduction without compromising safety.

Connecting Rod Weight Optimization

Observation:

- Fatigue safety factor = 6.36
- Life = 10^9 cycles (over-designed)

Suggested Improvement:

- Reduce beam cross-section thickness by 5–8%.
- Perform re-evaluation of fatigue safety.

Potential benefit:

- Reduced reciprocating mass
- Lower inertia forces
- Improved engine efficiency

Crankshaft Dynamic Enhancement

Observation:

- Large frequency separation (7.8×).

Suggested Improvement:

- Conduct harmonic response analysis.
- Evaluate torsional vibration under combustion pulse excitation.
- Optimize counterweight geometry for improved balance.

Potential benefit:

- Reduced vibration
- Improved bearing life
- Lower NVH levels

Thermal Performance Enhancement

Observation:

- Thermal expansion within acceptable range.

Suggested Improvement:

- Optimize cooling fin spacing.
- Investigate forced convection modelling.
- Consider transient startup thermal shock study.

Potential benefit:

- Improved heat dissipation
- Reduced thermal stress gradients
- Enhanced durability

Final Engineering Statement

While the current design demonstrates adequate safety margins under assumed operating conditions, optimization studies indicate opportunities for weight reduction, improved fatigue performance, and enhanced dynamic stability.

The current design demonstrates conservative safety margins across structural and fatigue domains, indicating potential for weight optimization in future design iterations without compromising reliability.

Future work involving nonlinear, transient, and harmonic analyses is recommended for production-level validation.